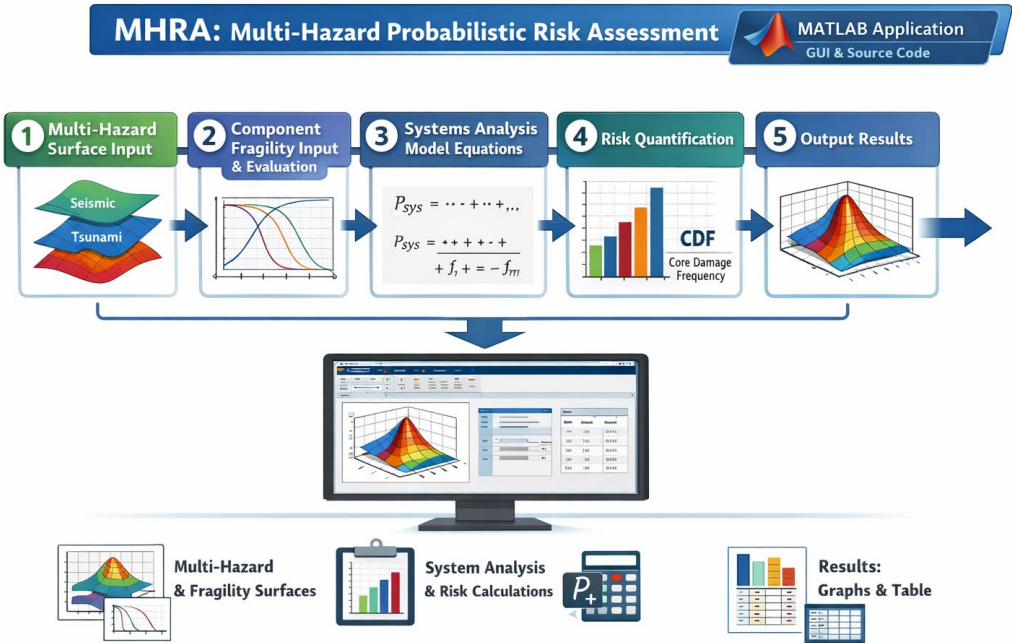
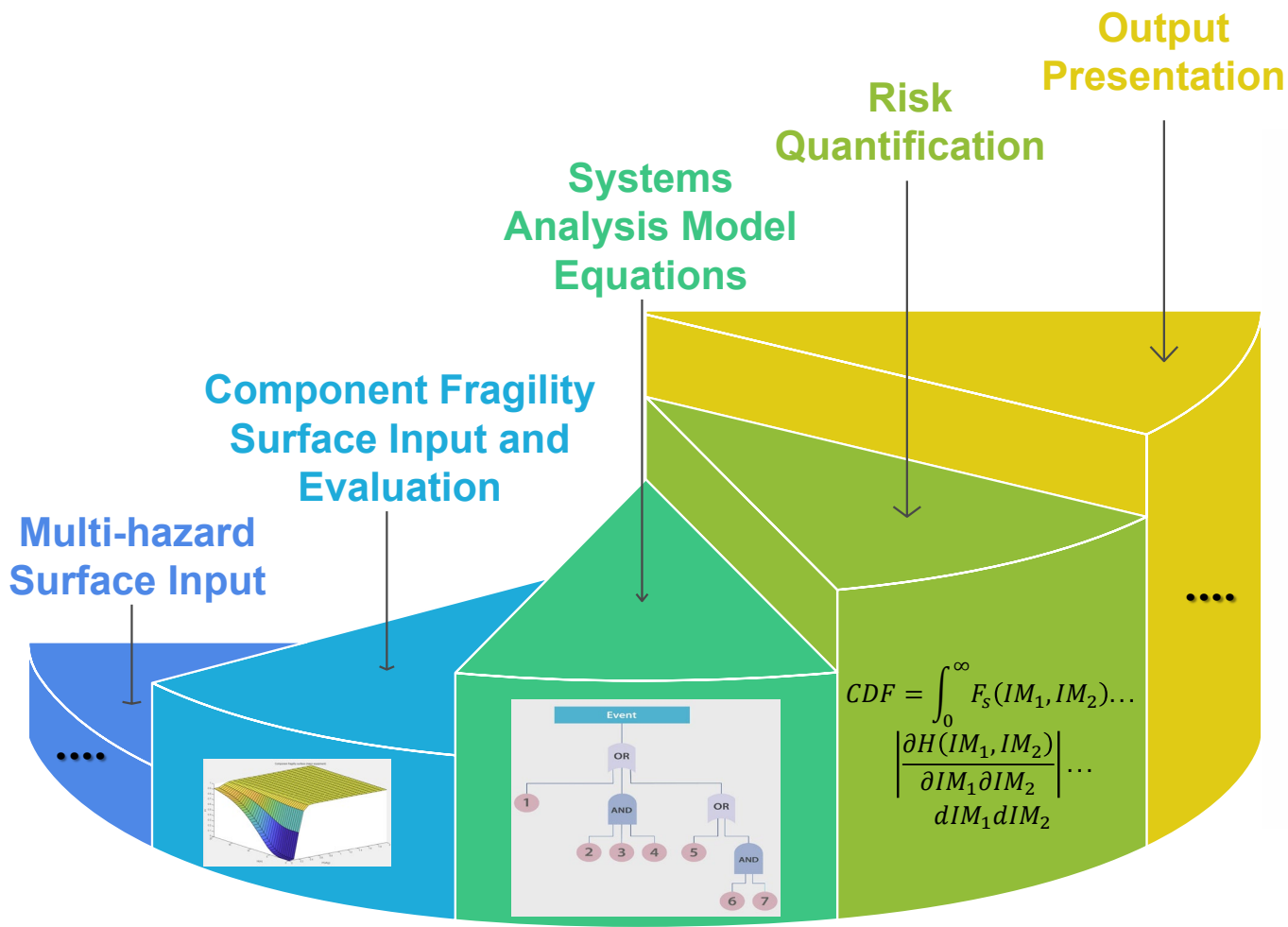
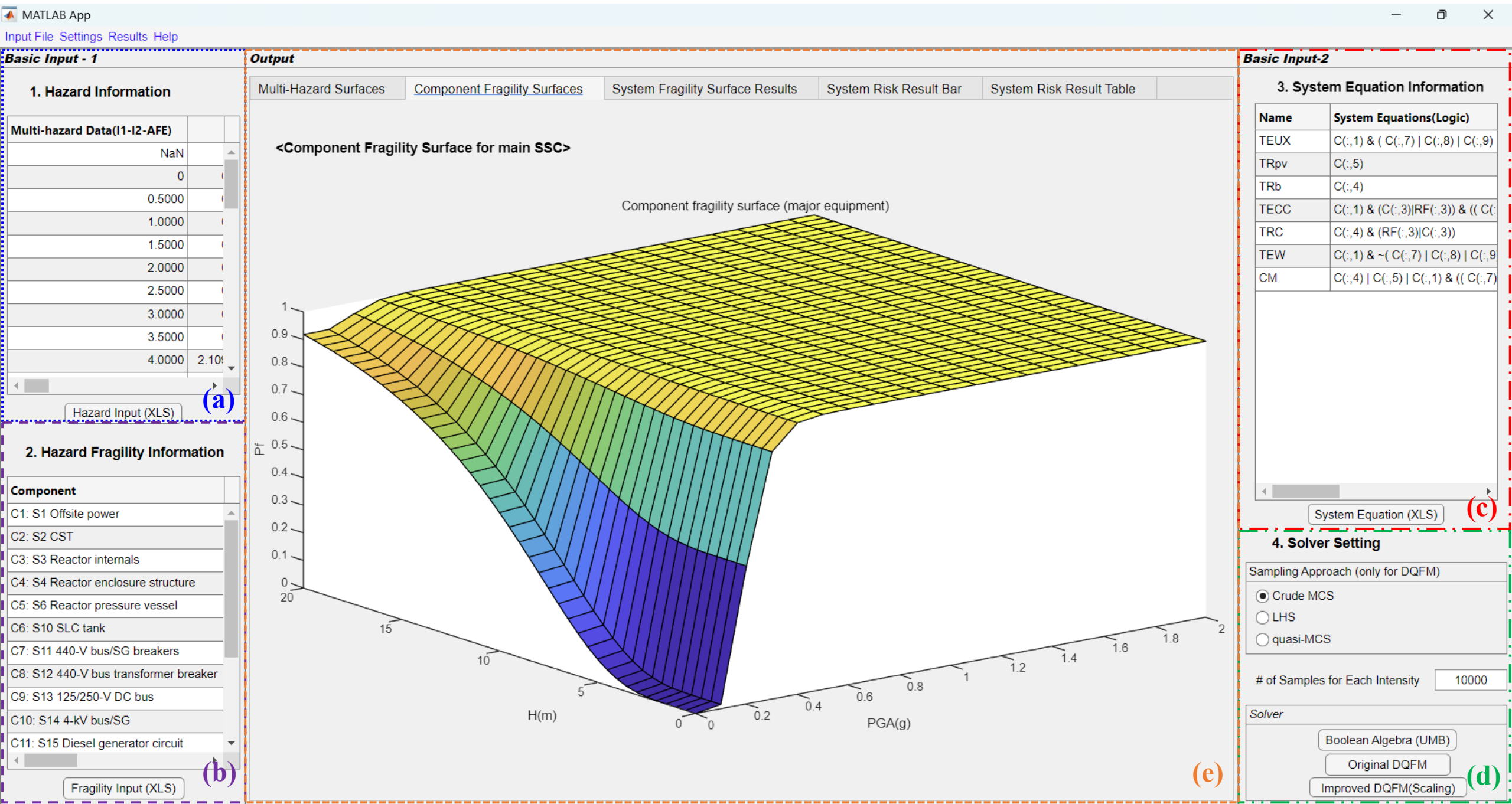


MHRA Program Modules



Matlab GUI Application Configuration



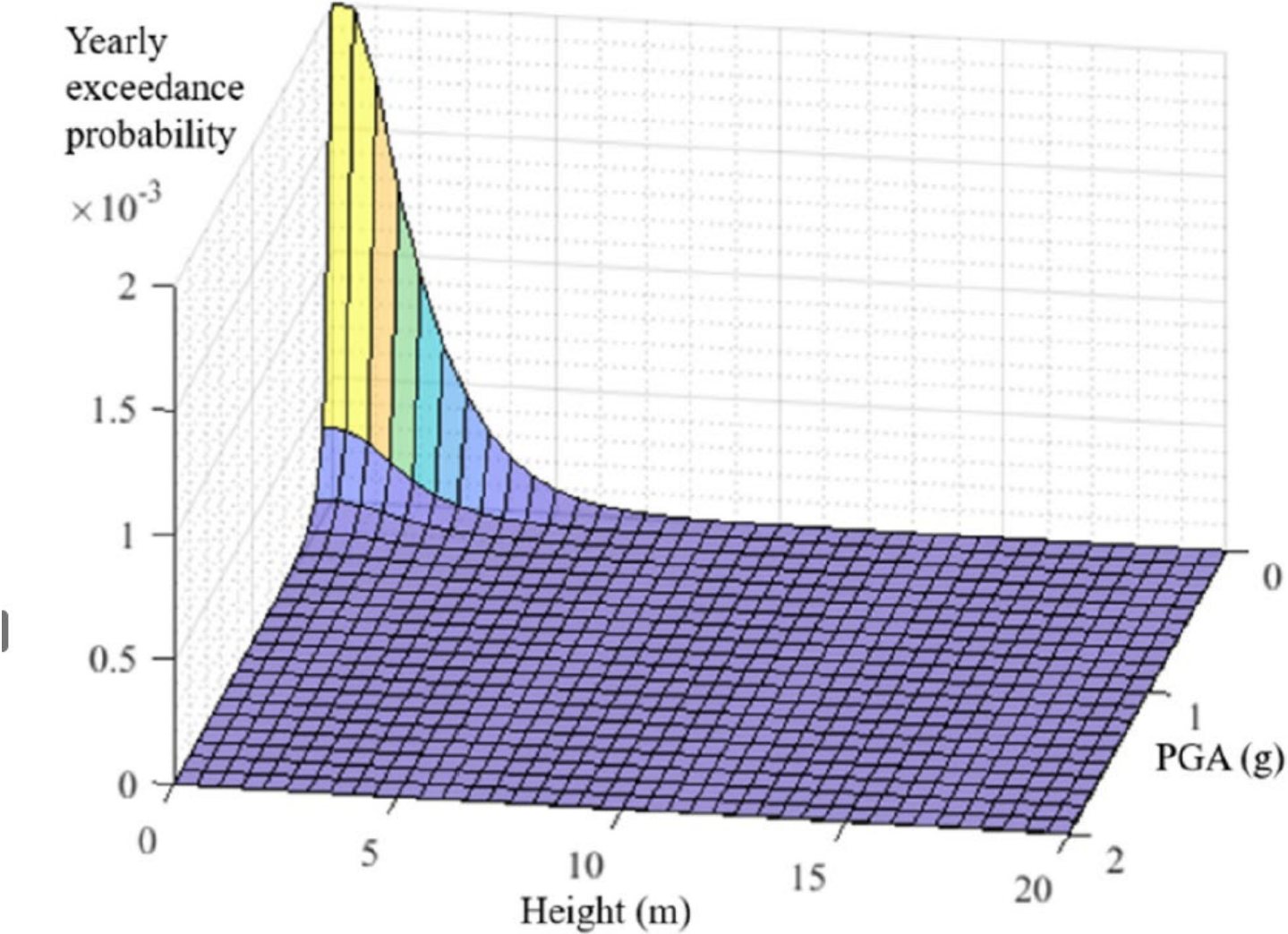


Fig. 6. Earthquake–tsunami hazard information for the numerical example (adopted from Refs. [19,23]).

Kwag, S., Ha, J. G., Kim, M. K., & Kim, J. H. (2019). Development of efficient external multi-hazard risk quantification methodology for nuclear facilities. *Energies*, 12(20), 3925.
Choi, E., Kwag, S., Ha, J. G., & Hahm, D. (2021). Development of a two-stage DQFM to improve efficiency of single-and multi-hazard risk quantification for nuclear facilities. *Energies*, 14(4), 1017.
Choi, E., Kwag, S., & Hahm, D. (2024). Multihazard capacity optimization of an NPP using a multi-objective genetic algorithm and sampling-based PSA. *Nuclear Engineering and Technology*, 56(2), 644-654.
Choi, E., Kwag, S., Kim, J. H., Ha, J. G., Hahm, D., & Kim, M. (2024). A review of COHRISK: Multihazard risk quantification software for nuclear power plants. *Nuclear Engineering and Technology*, 56(12), 5281-5290.

Table 1
Seismic fragility and random failure probability information of LGS NPP components (adapted from Ref. [19]).

Component		$R_{ms} (A_{ms})$	β_{Rcs}	B_{Ccs}	Mean failure rate (per yr)
S_1	Offsite power	0.20g	0.226	0.226	–
S_2	Condensate storage tank	0.24g	0.273	0.273	–
S_3	Reactor internals	0.67g	0.300	0.300	–
S_4	Reactor enclosure structure	1.05g	0.282	0.282	–
S_6	Reactor pressure vessel	1.25g	0.252	0.252	–
S_{10}	Standby liquid control system tank	1.33g	0.233	0.233	–
S_{11}	440-V bus/steam generator breakers	1.46g	0.411	0.411	–
S_{12}	440-V bus transformer breaker	1.49g	0.397	0.397	–
S_{13}	125/250-V DC bus	1.49g	0.397	0.397	–
S_{14}	4-kV bus/steam generator	1.49g	0.397	0.397	–
S_{15}	Diesel generator circuit	1.56g	0.368	0.368	–
S_{16}	Diesel generator heat and vent	1.55g	0.363	0.363	–
S_{17}	Residual heat removal system heat exchangers	1.09g	0.330	0.330	–
DG_R	DGR – diesel generator common mode	–	–	–	0.00125
W_R	WR – containment heat removal	–	–	–	0.00026
C_R	CR – scram system mechanical failure	–	–	–	1.00E-05
SLC_R	SLCR – standby liquid control	–	–	–	0.01

Table 2
Tsunami fragility of LGS NPP components (adapted from Ref. [19]).

Component		$R_{mt} (A_{mt})$	β_{Rct}	B_{Cct}
S_1	Offsite power	10 m	0.354	0.354
S_2	Condensate storage tank	10 m	0.212	0.212
S_{11}	440-V bus/SG breakers	11 m	0.212	0.212
S_{12}	440-V bus transformer breaker	11 m	0.212	0.212
S_{13}	125/250-V DC bus	11 m	0.212	0.212
S_{14}	4-kV bus/SG	11 m	0.212	0.212
S_{15}	Diesel generator circuit	11 m	0.212	0.212
S_{17}	RHR heat exchangers	10 m	0.212	0.212

Kwag, S., Ha, J. G., Kim, M. K., & Kim, J. H. (2019). Development of efficient external multi-hazard risk quantification methodology for nuclear facilities. *Energies*, 12(20), 3925.
Choi, E., Kwag, S., Ha, J. G., & Hahm, D. (2021). Development of a two-stage DQFM to improve efficiency of single-and multi-hazard risk quantification for nuclear facilities. *Energies*, 14(4), 1017.
Choi, E., Kwag, S., & Hahm, D. (2024). Multihazard capacity optimization of an NPP using a multi-objective genetic algorithm and sampling-based PSA. *Nuclear Engineering and Technology*, 56(2), 644-654.
Choi, E., Kwag, S., Kim, J. H., Ha, J. G., Hahm, D., & Kim, M. (2024). A review of COHRISK: Multihazard risk quantification software for nuclear power plants. *Nuclear Engineering and Technology*, 56(12), 5281-5290.

$$\left[\begin{array}{l} T_s E_s UX = (S_{1s} \cup S_{1t}) \cap A; \\ A = (S_{11s} \cup S_{11t}) \cup (S_{12s} \cup S_{12t}) \cup (S_{13s} \cup S_{13t}) \cup (S_{14s} \cup S_{14t}) \cup (S_{15s} \cup S_{15t}) \cup (S_{16s} \cup S_{16t}) \cup DG_R \\ \\ T_s R_b = (S_{4s} \cup S_{4t}) \\ T_s R_{pv} = (S_{6s} \cup S_{6t}) \\ \\ T_s E_s C_m C_2 = (S_{1s} \cup S_{1t}) \cap ((S_{3s} \cup S_{3t}) \cup C_R) \cap (A \cup (S_{10s} \cup S_{10t}) \cup SLC_R) \\ \\ T_s R_b C_m = (S_{4s} \cup S_{4t}) \cap (C_R \cup (S_{3s} \cup S_{3t})) \\ \\ T_s E_b W_m = (S_{1s} \cup S_{1t}) \cap \bar{A} \cap \left[\overline{(S_{17s} \cup S_{17t}) \cap W_R} \cup \overline{(S_{2s} \cup S_{2t}) \cap (S_{17s} \cup S_{17t})} \right] \end{array} \right]$$

Core Melt Damage Sequence:

$$CM = (S_{4s} \cup S_{4t}) \cup (S_{6s} \cup S_{6t}) \cup (S_{1s} \cup S_{1t}) \\ \cap \left\{ A \cup \left[(S_{3s} \cup S_{3t}) \cup C_R \right] \cap \left[(S_{10s} \cup S_{10t}) \cup SLC_R \right] \cup \left[(S_{17s} \cup S_{17t}) \cup W_R \right] \right\}$$

Kwag, S., Ha, J. G., Kim, M. K., & Kim, J. H. (2019). Development of efficient external multi-hazard risk quantification methodology for nuclear facilities. *Energies*, 12(20), 3925.
Choi, E., Kwag, S., Ha, J. G., & Hahm, D. (2021). Development of a two-stage DQFM to improve efficiency of single-and multi-hazard risk quantification for nuclear facilities. *Energies*, 14(4), 1017.
Choi, E., Kwag, S., & Hahm, D. (2024). Multihazard capacity optimization of an NPP using a multi-objective genetic algorithm and sampling-based PSA. *Nuclear Engineering and Technology*, 56(2), 644-654.
Choi, E., Kwag, S., Kim, J. H., Ha, J. G., Hahm, D., & Kim, M. (2024). A review of COHRISK: Multihazard risk quantification software for nuclear power plants. *Nuclear Engineering and Technology*, 56(12), 5281-5290.

1. Hazard Information

Multi-hazard Data(I1-I2-AFE)	
NaN	
0	
0.5000	
1.0000	
1.5000	
2.0000	
2.5000	
3.0000	
3.5000	
4.0000	2.109

Hazard Input (XLS)

2. Hazard Fragility Information

Component
C1: S1 Offsite power
C2: S2 CST
C3: S3 Reactor internals
C4: S4 Reactor enclosure structure
C5: S6 Reactor pressure vessel
C6: S10 SLC tank
C7: S11 440-V bus/SG breakers
C8: S12 440-V bus transformer breaker
C9: S13 125/250-V DC bus
C10: S14 4-kV bus/SG
C11: S15 Diesel generator circuit

Fragility Input (XLS)

Output

Multi-Hazard Surfaces

Component Fragility Surfaces

System Fragility Surface Results

System Risk Result Bar

System Risk Result Table

<Multi-Hazard Surface>

Multi-hazard surface

Multi-Hazard Surface Information

3. System Equation Information

Name	System Equations(Logic)
TEUX	C(:,1) & (C(:,7) C(:,8) C(:,9)
TRpv	C(:,5)
TRb	C(:,4)
TECC	C(:,1) & (C(:,3) RF(:,3)) & ((C(:,
TRC	C(:,4) & (RF(:,3) C(:,3))
TEW	C(:,1) & ~(C(:,7) C(:,8) C(:,9
CM	C(:,4) C(:,5) C(:,1) & ((C(:,7)

System Equation (XLS)

4. Solver Setting

Sampling Approach (only for DQFM)

☒ Crude MCS

☐ LHS

☐ quasi-MCS

of Samples for Each Intensity

10000

Solver

Boolean Algebra (UMB)

Original DQFM

Improved DQFM(Scaling)

1. Hazard Information

Multi-hazard Data(I1-I2-AFE)	
NaN	
0	
0.5000	
1.0000	
1.5000	
2.0000	
2.5000	
3.0000	
3.5000	
4.0000	2.10

Hazard Input (XLS)

2. Hazard Fragility Information

Component
C1: S1 Offsite power
C2: S2 CST
C3: S3 Reactor internals
C4: S4 Reactor enclosure structure
C5: S6 Reactor pressure vessel
C6: S10 SLC tank
C7: S11 440-V bus/SG breakers
C8: S12 440-V bus transformer breaker
C9: S13 125/250-V DC bus
C10: S14 4-kV bus/SG
C11: S15 Diesel generator circuit

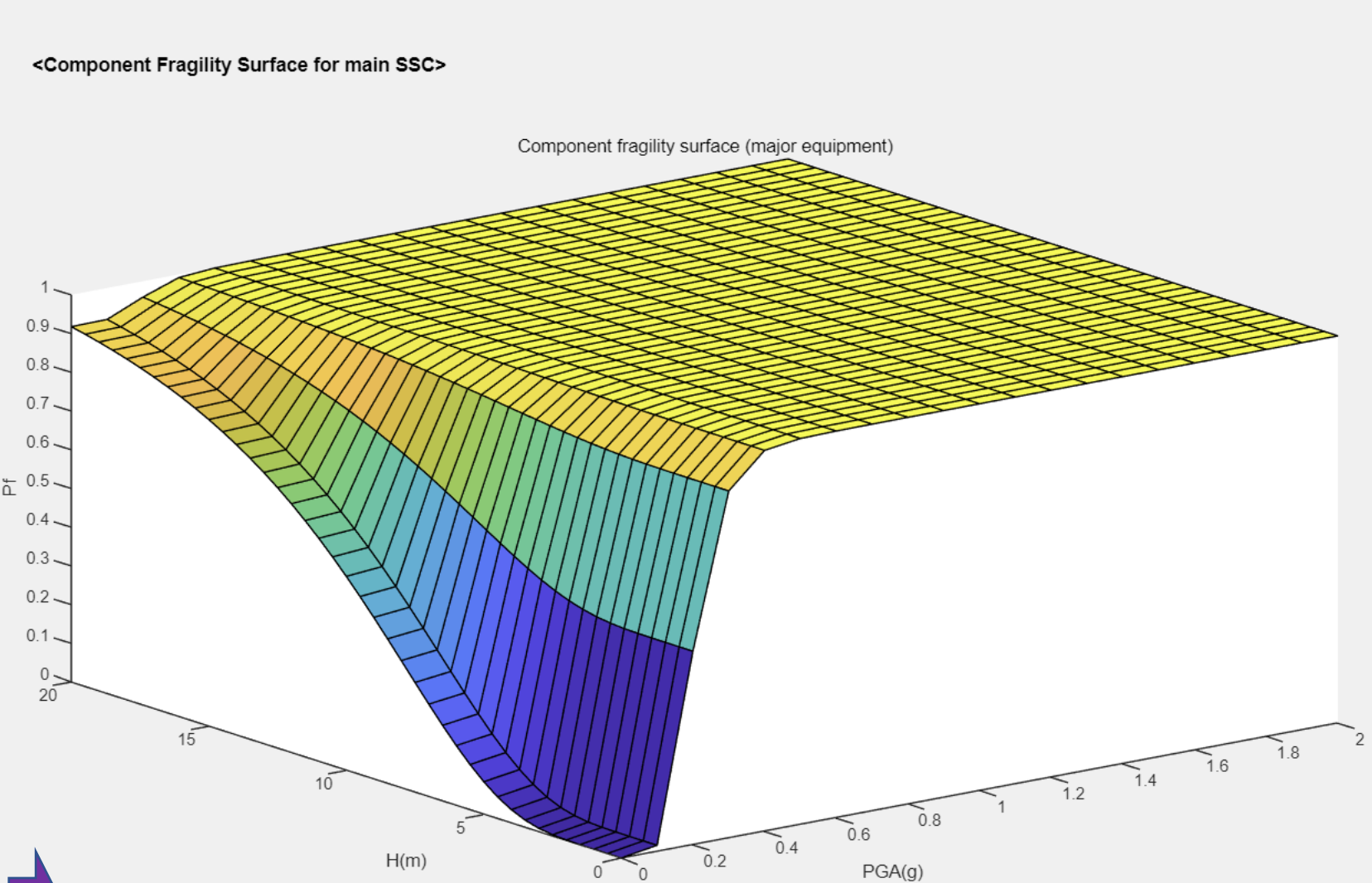
Fragility Input (XLS)

Output

Multi-Hazard SurfacesComponent Fragility SurfacesSystem Fragility Surface ResultsSystem Risk Result BarSystem Risk Result Table

<Component Fragility Surface for main SSC>

Component fragility surface (major equipment)



Basic Input-2

3. System Equation Information

Name	System Equations(Logic)
TEUX	C(:,1) & (C(:,7) C(:,8) C(:,9)
TRpv	C(:,5)
TRb	C(:,4)
TECC	C(:,1) & (C(:,3) RF(:,3)) & ((C(:,
TRC	C(:,4) & (RF(:,3) C(:,3))
TEW	C(:,1) & ~(C(:,7) C(:,8) C(:,9
CM	C(:,4) C(:,5) C(:,1) & ((C(:,7)

System Equation (XLS)

4. Solver Setting

Sampling Approach (only for DQFM)

☒ Crude MCS

☐ LHS

☐ quasi-MCS

of Samples for Each Intensity10000

Solver

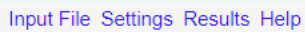
Boolean Algebra (UMB)

Original DQFM

Improved DQFM(Scaling)

Component Multi-Hazard Fragility Surfaces for LGS NPP SSCs

Systems Analysis Model Input & System Multi-Hazard Fragility Surface Results



Output

Multi-Hazard Surfaces

Component Fragility Surfaces

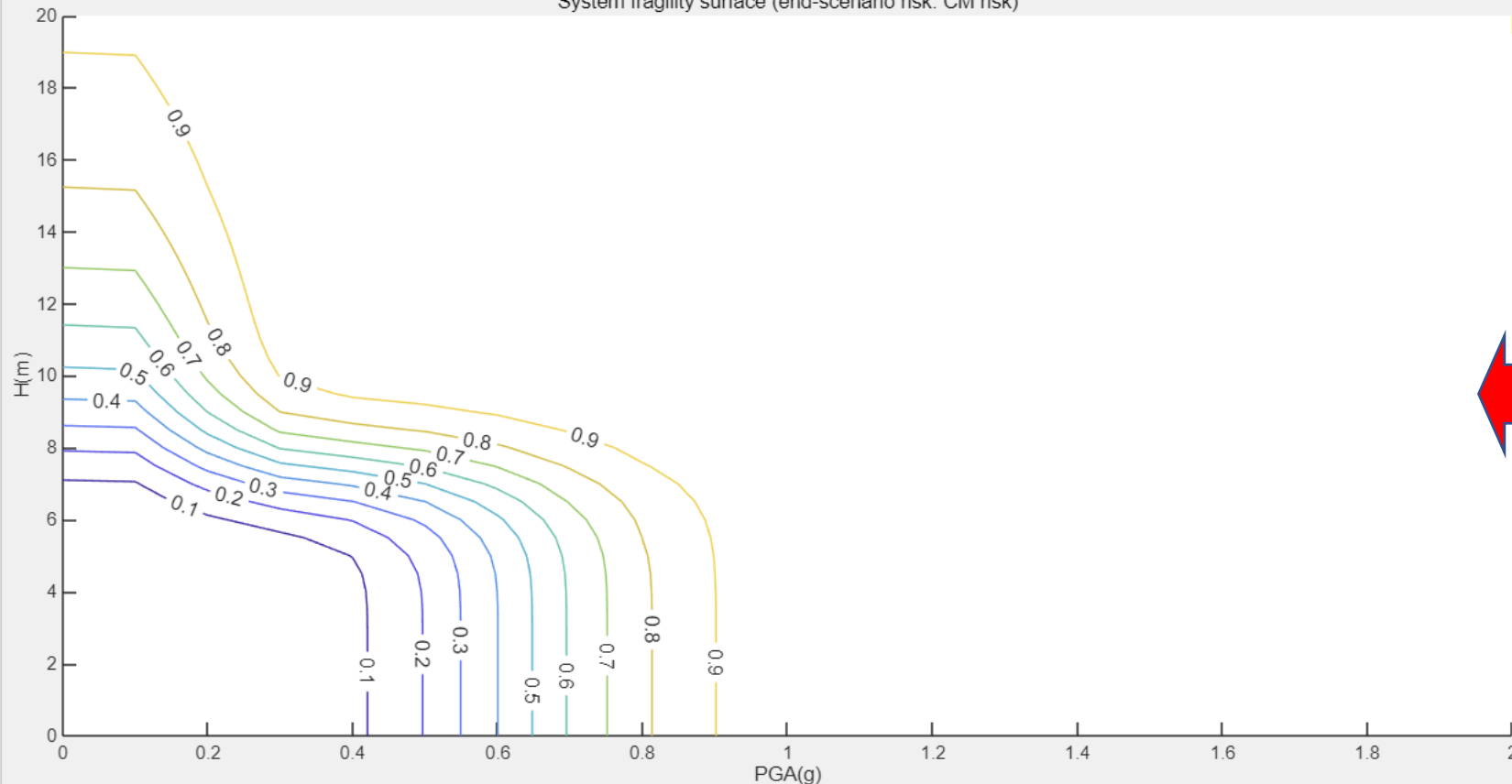
System Fragility Surface Results

System Risk Result Bar

System Risk Result Table

<System Fragility Surface Results>

System fragility surface (end-scenario risk: CM risk)



Basic Input-2

Name	System Equations(Logic)
TEUX	$C(:,1) \& (C(:,7) C(:,8) C(:,9))$
TRpv	$C(:,5)$
TRb	$C(:,4)$
TECC	$C(:,1) \& (C(:,3) RF(:,3)) \& ((C(:,4) C(:,5) C(:,6) C(:,7) C(:,8) C(:,9)) \& TRb)$
TRC	$C(:,4) \& (RF(:,3) C(:,3))$
TEW	$C(:,1) \& \sim (C(:,7) C(:,8) C(:,9))$
CM	$C(:,4) C(:,5) C(:,1) \& ((C(:,7) C(:,8) C(:,9)) \& TRb)$

System Equation (XLS)

Sampling Approach (only for DQFM)

- ☒ Crude MCS
- ☐ LHS
- ☐ quasi-MCS

# of Samples for Each Intensity	10000
---------------------------------	-------

Solver

Boolean Algebra (UMB)

Original DQFM

Improved DQFM(Scaling)

1. Hazard Information

Multi-hazard Data(I1-I2-AFE)	
NaN	
0	
0.5000	
1.0000	
1.5000	
2.0000	
2.5000	
3.0000	
3.5000	
4.0000	2.10e-05

Hazard Input (XLS)

2. Hazard Fragility Information

Component
C1: S1 Offsite power
C2: S2 CST
C3: S3 Reactor internals
C4: S4 Reactor enclosure structure
C5: S6 Reactor pressure vessel
C6: S10 SLC tank
C7: S11 440-V bus/SG breakers
C8: S12 440-V bus transformer breaker
C9: S13 125/250-V DC bus
C10: S14 4-kV bus/SG
C11: S15 Diesel generator circuit

Fragility Input (XLS)

Multi-Hazard Surfaces

Component Fragility Surfaces

System Fragility Surface Results

System Risk Result Bar

System Risk Result Table

<System Risk Result Bar>

System Equations	Risk (/yr) × 10 ⁻⁵
TEUX	7.3966e-06
TRpv	4.0996e-07
TRb	1.0397e-06
TECC	1.4091e-06
TRC	5.833e-07
TEW	7.135e-07
CM	9.6242e-06

3. System Equation Information

Name	System Equations(Logic)
TEUX	C(:,1) & (C(:,7) C(:,8) C(:,9)
TRpv	C(:,5)
TRb	C(:,4)
TECC	C(:,1) & (C(:,3)) RF(:,3)) & ((C(:,
TRC	C(:,4) & (RF(:,3)) C(:,3))
TEW	C(:,1) & ~(C(:,7) C(:,8) C(:,9
CM	C(:,4) C(:,5) C(:,1) & ((C(:,7)

System Equation (XLS)

4. Solver Setting

Sampling Approach (only for DQFM)

☒ Crude MCS

☐ LHS

☐ quasi-MCS

of Samples for Each Intensity

10000

Solver

Boolean Algebra (UMB)

Original DQFM

Improved DQFM(Scaling)

Risk Quantification Bar Result using Three Different Solvers and Three Different Samplers

1. Hazard Information

Multi-hazard Data(I1-I2-AFE)	
NaN	
0	
0.5000	
1.0000	
1.5000	
2.0000	
2.5000	
3.0000	
3.5000	
4.0000	2.10

Hazard Input (XLS)

2. Hazard Fragility Information

Component
C1: S1 Offsite power
C2: S2 CST
C3: S3 Reactor internals
C4: S4 Reactor enclosure structure
C5: S6 Reactor pressure vessel
C6: S10 SLC tank
C7: S11 440-V bus/SG breakers
C8: S12 440-V bus transformer breaker
C9: S13 125/250-V DC bus
C10: S14 4-kV bus/SG
C11: S15 Diesel generator circuit

Fragility Input (XLS)

Multi-Hazard Surfaces

Component Fragility Surfaces

System Fragility Surface Results

System Risk Result Bar

System Risk Result Table

<System Risk Result Table>

Name	Risk(/yr)	CDF contribution(/yr)
TEUX	7.3966e-06	7.3966e-06
TRpv	4.0996e-07	4.0996e-07
TRb	1.0397e-06	1.0397e-06
TECC	1.4091e-06	1.4091e-06
TRC	5.8330e-07	5.8330e-07
TEW	7.1350e-07	7.1350e-07
CM	9.6242e-06	9.6242e-06

3. System Equation Information

Name	System Equations(Logic)
TEUX	C(:,1) & (C(:,7) C(:,8) C(:,9)
TRpv	C(:,5)
TRb	C(:,4)
TECC	C(:,1) & (C(:,3) RF(:,3)) & ((C(:,
TRC	C(:,4) & (RF(:,3) C(:,3))
TEW	C(:,1) & ~(C(:,7) C(:,8) C(:,9
CM	C(:,4) C(:,5) C(:,1) & ((C(:,7)

System Equation (XLS)

4. Solver Setting

Sampling Approach (only for DQFM)

☒ Crude MCS

☐ LHS

☐ quasi-MCS

of Samples for Each Intensity

10000

Solver

Boolean Algebra (UMB)

Original DQFM

Improved DQFM(Scaling)

Risk Quantification Bar Result using Three Different Solvers and Three Different Samplers

Appendix. File Type and Import for Example Case

MATLAB App

Input File Settings Results Help

Basic Input - 1

1. Hazard Information

Multi-hazard Data(I1-I2-AFE)

NaN	
0	
0.5000	
1.0000	
1.5000	
2.0000	
2.5000	
3.0000	
3.5000	
4.0000	2.10

Hazard Input (XLS)

2. Hazard Fragility Information

Component

C1: S1 Offsite power
C2: S2 CST
C3: S3 Reactor internals
C4: S4 Reactor enclosure structure
C5: S6 Reactor pressure vessel
C6: S10 SLC tank
C7: S11 440-V bus/SG breakers
C8: S12 440-V bus transformer breaker
C9: S13 125/250-V DC bus
C10: S14 4-kV bus/SG
C11: S15 Diesel generator circuit

Fragility Input (XLS)



Input_Data_Benchmark.xlsx

Button Click and Import Sheet1 of the EXCEL file.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V		
1		PGA(g)																						
2	H(m)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2		
3		0	2.00E-03	0.00035	0.00012	4.44E-05	1.87E-05	8.62E-06	4.29E-06	2.26E-06	1.26E-06	7.26E-07	4.35E-07	2.69E-07	1.71E-07	1.11E-07	7.36E-08	4.98E-08	3.42E-08	2.39E-08	1.69E-08	1.21E-08	8.81E-09	
4	5.00E-01	1.99E-03	0.00035	0.00012	4.42E-05	1.86E-05	8.59E-06	4.27E-06	2.26E-06	1.25E-06	7.25E-07	4.34E-07	2.68E-07	1.70E-07	1.11E-07	7.34E-08	4.97E-08	3.42E-08	2.38E-08	1.69E-08	1.21E-08	8.79E-09		
5	1.00E+00	1.70E-03	0.0003	0.0001	3.98E-05	1.69E-05	7.87E-06	3.93E-06	2.09E-06	1.16E-06	6.75E-07	4.06E-07	2.51E-07	1.60E-07	1.04E-07	6.92E-08	4.68E-08	3.23E-08	2.26E-08	1.60E-08	1.15E-08	8.35E-09		
6	1.50E+00	1.27E-03	0.00022	8.07E-05	3.14E-05	1.36E-05	6.38E-06	3.22E-06	1.72E-06	9.65E-07	5.63E-07	3.40E-07	2.11E-07	1.35E-07	8.82E-08	5.88E-08	3.99E-08	2.76E-08	1.93E-08	1.37E-08	9.88E-09	7.19E-09		
7	2.00E+00	9.19E-04	0.00016	5.78E-05	2.29E-05	1.00E-05	4.75E-06	2.41E-06	1.30E-06	7.31E-07	4.29E-07	2.60E-07	1.62E-07	1.04E-07	6.80E-08	4.55E-08	3.09E-08	2.14E-08	1.50E-08	1.07E-08	7.72E-09	5.63E-09		
8	2.5	6.53E-04	0.0001	3.90E-05	1.56E-05	6.89E-06	3.29E-06	1.68E-06	9.09E-07	5.14E-07	3.02E-07	1.84E-07	1.15E-07	7.37E-08	4.84E-08	3.24E-08	2.21E-08	1.53E-08	1.08E-08	7.68E-09	5.54E-09	4.04E-09		
9	3	4.55E-04	6.56E-05	2.51E-05	1.02E-05	4.50E-06	2.16E-06	1.11E-06	6.00E-07	3.40E-07	2.00E-07	1.22E-07	7.65E-08	4.92E-08	3.23E-08	2.17E-08	1.48E-08	1.03E-08	7.23E-09	5.16E-09	3.72E-09	2.72E-09		
10	3.5	0.00031	4.04E-05	1.56E-05	6.36E-06	2.83E-06	1.36E-06	7.01E-07	3.80E-07	2.16E-07	1.27E-07	7.77E-08	4.87E-08	3.14E-08	2.06E-08	1.38E-08	9.46E-09	6.57E-09	4.63E-09	3.30E-09	2.39E-09	1.75E-09		
11	4	0.00021	2.44E-05	9.50E-06	3.88E-06	1.73E-06	8.36E-07	4.31E-07	2.34E-07	1.33E-07	7.86E-08	4.80E-08	3.01E-08	1.94E-08	1.28E-08	8.58E-09	5.86E-09	4.07E-09	2.87E-09	2.05E-09	1.48E-09	1.08E-09		
12	4.5	0.00014	1.46E-05	5.69E-06	2.33E-06	1.04E-06	5.04E-07	2.60E-07	1.41E-07	8.05E-08	4.76E-08	2.91E-08	1.83E-08	1.18E-08	7.75E-09	5.21E-09	3.56E-09	2.47E-09	1.74E-09	1.25E-09	9.02E-10	6.60E-10		
13	5	9.35E-05	8.61E-06	3.38E-06	1.39E-06	6.21E-07	3.01E-07	1.55E-07	8.46E-08	4.82E-08	2.85E-08	1.74E-08	1.09E-08	7.05E-09	4.65E-09	3.12E-09	2.14E-09	1.49E-09	1.05E-09	7.49E-10	5.42E-10	3.96E-10		
14	5.5	6.17E-05	5.08E-06	2.00E-06	8.22E-07	3.69E-07	1.79E-07	9.23E-08	5.03E-08	2.87E-08	1.70E-08	1.04E-08	6.52E-09	4.20E-09	2.77E-09	1.86E-09	1.27E-09	8.86E-10	6.25E-10	4.47E-10	3.23E-10	2.37E-10		
15	6	4.06E-05	3.00E-06	1.18E-06	4.87E-07	2.19E-07	1.06E-07	5.48E-08	2.99E-08	1.70E-08	1.01E-08	6.16E-09	3.88E-09	2.50E-09	1.65E-09	1.11E-09	7.59E-10	5.28E-10	3.72E-10	2.66E-10	1.93E-10	1.41E-10		
16	6.5	2.66E-05	1.77E-06	7.00E-07	2.89E-07	1.30E-07	6.30E-08	3.26E-08	1.78E-08	1.01E-08	6.00E-09	3.67E-09	2.31E-09	1.49E-09	9.82E-10	6.60E-10	4.52E-10	3.14E-10	2.22E-10	1.59E-10	1.15E-10	8.40E-11		
17	7	1.75E-05	1.05E-06	4.17E-07	1.72E-07	7.73E-08	3.75E-08	1.94E-08	1.06E-08	6.04E-09	3.58E-09	2.19E-09	1.38E-09	8.89E-10	5.87E-10	3.95E-10	2.70E-10	1.88E-10	1.33E-10	9.48E-11	6.86E-11	5.02E-11		
18	7.5	1.15E-05	6.30E-07	2.49E-07	1.03E-07	4.63E-08	2.25E-08	1.16E-08	6.35E-09	3.62E-09	2.15E-09	1.31E-09	8.27E-10	5.34E-10	3.52E-10	2.37E-10	1.62E-10	1.13E-10	7.96E-11	5.69E-11	4.12E-11	3.02E-11		
19	8	7.60E-06	3.79E-07	1.50E-07	6.20E-08	2.79E-08	1.35E-08	7.01E-09	3.83E-09	2.18E-09	1.29E-09	7.92E-10	4.99E-10	3.22E-10	2.12E-10	1.43E-10	9.78E-11	6.80E-11	4.80E-11	3.43E-11	2.49E-11	1.82E-11		
20	8.5	5.03E-06	2.29E-07	9.08E-08	3.76E-08	1.69E-08	8.22E-09	4.25E-09	2.32E-09	1.33E-09	7.85E-10	4.81E-10	3.03E-10	1.95E-10	1.29E-10	8.67E-11	5.94E-11	4.13E-11	2.92E-11	2.09E-11	1.51E-11	1.11E-11		
21	9	3.34E-06	1.40E-07	5.54E-08	2.29E-08	1.03E-08	5.01E-09	2.60E-09	1.42E-09	8.09E-10	4.80E-10	2.94E-10	1.85E-10	1.19E-10	7.87E-11	5.30E-11	3.63E-11	2.52E-11	1.78E-11	1.27E-11	9.23E-12	6.76E-12		
22	9.5	2.23E-06	8.57E-08	3.40E-08	1.41E-08	6.34E-09	3.08E-09	1.60E-09	8.72E-10	4.98E-10	2.95E-10	1.81E-10	1.14E-10	7.34E-11	4.84E-11	3.26E-11	2.23E-11	1.55E-11	1.10E-11	7.84E-12	5.68E-12	4.16E-12		
23	10	1.49E-06	5.30E-08	2.10E-08	8.71E-09	3.93E-09	1.91E-09	9.89E-10	5.40E-10	3.08E-10	1.83E-10	1.12E-10	7.04E-11	4.55E-11	3.00E-11	2.02E-11	1.38E-11	9.62E-12	6.79E-12	4.86E-12	3.52E-12	2.58E-12		
24	10.5	1.00E-06	3.30E-08	1.31E-08	5.43E-09	2.45E-09	1.19E-09	6.17E-10	3.37E-10	1.92E-10	1.14E-10	6.98E-11	4.40E-11	2.84E-11	1.87E-11	1.26E-11	8.63E-12	6.01E-12	4.24E-12	3.03E-12	2.20E-12	1.61E-12		
25	11	6.76E-07	2.07E-08	8.23E-09	3.41E-09	1.54E-09	7.47E-10	3.87E-10	2.12E-10	1.21E-10	7.16E-11	4.38E-11	2.76E-11	1.78E-11	1.18E-11	7.92E-12	5.42E-12	3.77E-12	2.66E-12	1.91E-12	1.38E-12	1.01E-12		
26	11.5	4.59E-07	1.31E-08	5.21E-09	2.16E-09	9.72E-10	4.73E-10	2.45E-10	1.34E-10	7.64E-11	4.53E-11	2.77E-11	1.75E-11	1.13E-11	7.45E-12	5.01E-12	3.43E-12	2.39E-12	1.69E-12	1.21E-12	8.74E-13	6.40E-13		
27	12	3.13E-07	8.34E-09	3.31E-09	1.37E-09	6.19E-10	3.01E-10	1.56E-10	8.53E-11	4.87E-11	2.89E-11	1.77E-11	1.11E-11	7.19E-12	4.75E-12	3.19E-12	2.19E-12	1.52E-12	1.07E-12	7.69E-13	5.57E-13	4.08E-13		
28	12.5	2.14E-07	5.34E-09	2.13E-09	8.81E-10	3.97E-10	1.93E-10	1.00E-10	5.47E-11	3.12E-11	1.85E-11	1.13E-11	7.14E-12	4.61E-12	3.04E-12	2.05E-12	1.40E-12	9.77E-13	6.90E-13	4.93E-13	3.57E-13	2.62E-13		
29	13	1.47E-07	3.45E-09	1.37E-09	5.69E-10	2.56E-10	1.25E-10	6.47E-11	3.53E-11	2.02E-11	1.20E-11	7.33E-12	4.62E-12	2.98E-12	1.97E-12	1.32E-12	9.07E-13	6.31E-13	4.46E-13	3.19E-13	2.31E-13	1.69E-13		
30	13.5	1.02E-07	2.24E-09	8.92E-10	3.70E-10	1.67E-10	8.11E-11	4.20E-11	2.30E-11	1.31E-11	7.78E-12	4.76E-12	3.00E-12	1.94E-12	1.28E-12	8.61E-13	5.90E-13	4.10E-13	2.90E-13	2.07E-13	1.50E-13	1.10E-13		
31	14	7.07E-08	1.47E-09	5.83E-10	2.42E-10	1.09E-10	5.31E-11	2.75E-11	1.50E-11	8.58E-12	5.09E-12	3.12E-12	1.96E-12	1.27E-12	8.37E-13	5.63E-13	3.86E-13	2.69E-13	1.90E-13	1.36E-13	9.83E-14	7.20E-14		
32	14.5	4.93E-08	9.65E-10	3.84E-10	1.59E-10	7.18E-11	3.49E-11	1.81E-11	9.90E-12	5.65E-12	3.35E-12	2.05E-12	1.29E-12	8.35E-13	5.51E-13	3.71E-13	2.54E-13	1.77E-13	1.25E-13	8.94E-14	6.47E-14	4.74E-14		
33	15	3.45E-08	6.39E-10	2.54E-10	1.05E-10	4.76E-11	2.31E-11	1.20E-11	6.56E-12	3.74E-12	2.22E-12	1.36E-12	8.57E-13	5.53E-13	3.65E-13	2.46E-13	1.68E-13	1.17E-13	8.28E-14	5.92E-14	4.29E-14	3.14E-14		
34	15.5	2.43E-08	4.26E-10	1.69E-10	7.03E-11	3.17E-11	1.54E-11	8.00E-12	4.37E-12	2.50E-12	1.48E-12	9.07E-13	5.71E-13	3.69E-13	2.43E-13	1.64E-13	1.12E-13	7.81E-14	5.52E-14	3.95E-14	2.86E-14	2.09E-14		
35	16	1.72E-08	2.85E-10	1.14E-10	4.71E-11	2.13E-11	1.03E-11	5.36E-12	2.93E-12	1.67E-12	9.92E-13	6.08E-13	3.83E-13	2.47E-13	1.63E-13	1.10E-13	7.53E-14	5.24E-14	3.70E-14	2.65E-14	1.92E-14	1.40E-14		
36	16.5	1.22E-08	1.92E-10	7.66E-11	3.18E-11	1.43E-11	6.97E-12	3.61E-12	1.98E-12	1.13E-12	6.69E-13	4.10E-13	2.58E-13	1.67E-13	1.10E-13	7.41E-14	5.08E-14	3.53E-14	2.49E-14	1.79E-14	1.29E-14	9.47E-15		
37	17	8.67E-09	1.30E-10	5.19E-11	2.15E-11	9.71E-12	4.73E-12	2.45E-12	1.34E-12	7.65E-13	4.54E-13	2.78E-13	1.75E-13	1.13E-13	7.46E-14	5.02E-14	3.44E-14	2.40E-14	1.69E-14	1.21E-14	8.77E-15	6.42E-15		
38	17.5	6.20E-09	8.88E-11	3.54E-11	1.47E-11	6.62E-12	3.22E-12	1.67E-12	9.13E-13	5.21E-13	3.09E-13	1.89E-13	1.19E-13	7.71E-14	5.09E-14	3.42E-14	2.35E-14	1.63E-14	1.15E-14	8.25E-15	5.98E-15	4.38E-15		
39	18	4.46E-09	6.08E-11	2.42E-11	1.00E-11	4.53E-12	2.21E-12	1.14E-12	6.25E-13	3.57E-13	2.12E-13	1.30E												

Appendix. File Type and Import

MATLAB App

Input File

Settings

Results

Help

Basic Input - 1

1. Hazard Information

Multi-hazard Data(I1-I2-AFE)

NaN

0

0.5000

1.0000

1.5000

2.0000

2.5000

3.0000

3.5000

4.0000

2.1000

Hazard Input (XLS)

2. Hazard Fragility Information

Component

C1: S1 Offsite power

C2: S2 CST

C3: S3 Reactor internals

C4: S4 Reactor enclosure structure

C5: S6 Reactor pressure vessel

C6: S10 SLC tank

C7: S11 440-V bus/SG breakers

C8: S12 440-V bus transformer breaker

C9: S13 125/250-V DC bus

C10: S14 4-kV bus/SG

C11: S15 Diesel generator circuit

Fragility Input (XLS)

Input_Data_Benchmark.xlsx

	A	B	C	D	E	F	G	H
1		PGA			H			
2		Am	br	bu	Hm	br	bu	random_failure
3	C1: S1 Offsite power	0.2	0.2	0.25	10	0.5	0	0
4	C2: S2 CST	0.24	0.23	0.31	10	0.3	0	0
5	C3: S3 Reactor internals	0.67	0.28	0.32	1E+10	0	0	0
6	C4: S4 Reactor enclosure structure	1.05	0.31	0.25	1E+10	0	0	0
7	C5: S6 Reactor pressure vessel	1.25	0.28	0.22	1E+10	0	0	0
8	C6: S10 SLC tank	1.33	0.27	0.19	1E+10	0	0	0
9	C7: S11 440-V bus/SG breakers	1.46	0.38	0.44	11	0.3	0	0
10	C8: S12 440-V bus transformer breaker	1.49	0.36	0.43	11	0.3	0	0
11	C9: S13 125/250-V DC bus	1.49	0.36	0.43	11	0.3	0	0
12	C10: S14 4-kV bus/SG	1.49	0.36	0.43	11	0.3	0	0
13	C11: S15 Diesel generator circuit	1.56	0.32	0.41	11	0.3	0	0
14	C12: S16 Diesel generator heat and vent	1.55	0.28	0.43	1E+10	0	0	0
15	C13: S17 RHR heat exchangers	1.09	0.32	0.34	10	0.3	0	0
16	RF1: DGR Diesel generator common mod	0	0	0	0	0	0	0.00125
17	RF2: WR Containment heat removal	0	0	0	0	0	0	0.00026
18	RF3: CR Scram system mechanical failure	0	0	0	0	0	0	1.00E-05
19	RF4: SLCR Standby liquid control	0	0	0	0	0	0	0.01

Button Click
and Import
Sheet2 of
the EXCEL
file.

Appendix. File Type and Import



Input_Data_Benchmark.xlsx

Basic Input-2

3. System Equation Information

Name	System Equations(Logic)
TEUX	$C(:,1) \& (C(:,7) C(:,8) C(:,9))$
TRpv	$C(:,5)$
TRb	$C(:,4)$
TECC	$C(:,1) \& (C(:,3) RF(:,3)) \& ((C(:,7) C(:,8) C(:,9) C(:,10) C(:,11) C(:,12) RF(:,1)) C(:,6) RF(:,4))$
TRC	$C(:,4) \& (RF(:,3) C(:,3))$
TEW	$C(:,1) \& \sim(C(:,7) C(:,8) C(:,9) C(:,10) C(:,11) C(:,12) RF(:,1)) \& ((\sim C(:,13) \& RF(:,2)) (\sim C(:,2) \& C(:,13)))$
CM	$C(:,4) C(:,5) C(:,1) \& ((C(:,7) C(:,8) C(:,9) C(:,10) C(:,11) C(:,12) RF(:,1)) (C(:,3) RF(:,3))\&(C(:,6) RF(:,4)) (C(:,13) RF(:,2)))$

System Equation (XLS)

4. Solver Setting

Sampling Approach (only for DQFM)

☒ Crude MCS

☐ LHS

☐ quasi-MCS

of Samples for Each Intensity10000

Solver

Boolean Algebra (UMB)

Original DQFM

Improved DQFM(Scaling)

	A	B
1	Name	System Equations (Logic Tree)
2	TEUX	$C(:,1) \& (C(:,7) C(:,8) C(:,9) C(:,10) C(:,11) C(:,12) RF(:,1))$
3	TRpv	$C(:,5)$
4	TRb	$C(:,4)$
5	TECC	$C(:,1) \& (C(:,3) RF(:,3)) \& ((C(:,7) C(:,8) C(:,9) C(:,10) C(:,11) C(:,12) RF(:,1)) C(:,6) RF(:,4))$
6	TRC	$C(:,4) \& (RF(:,3) C(:,3))$
7	TEW	$C(:,1) \& \sim(C(:,7) C(:,8) C(:,9) C(:,10) C(:,11) C(:,12) RF(:,1)) \& ((\sim C(:,13) \& RF(:,2)) (\sim C(:,2) \& C(:,13)))$
8	CM	$C(:,4) C(:,5) C(:,1) \& ((C(:,7) C(:,8) C(:,9) C(:,10) C(:,11) C(:,12) RF(:,1)) (C(:,3) RF(:,3))\&(C(:,6) RF(:,4)) (C(:,13) RF(:,2)))$

“Event Tree and Fault Tree-type” Model -> “Logical Expression”

⋮

	C	D
1	System Equations (UMB)	Secondary ET
2	$C(:,1) \& (1 - (1 - C(:,7))) \& (1 - C(:,8)) \& (1 - C(:,9)) \& (1 - C(:,10)) \& (1 - C(:,11)) \& (1 - C(:,12)) \& (1 - RF(:,1))$	1
3	$C(:,5)$	1
4	$C(:,4)$	1
5	$C(:,1) \& (1 - (1 - C(:,3))) \& (1 - RF(:,3)) \& (1 - (1 - (1 - C(:,7))) \& (1 - C(:,8)) \& (1 - C(:,9)) \& (1 - C(:,10)) \& (1 - C(:,11)) \& (1 - C(:,12)) \& (1 - RF(:,1))) \& (1 - C(:,6)) \& (1 - RF(:,4))$	1
6	$C(:,4) \& (1 - (1 - RF(:,3)) \& (1 - C(:,3)))$	1
7	$C(:,1) \& (1 - (1 - (1 - C(:,7))) \& (1 - C(:,8)) \& (1 - C(:,9)) \& (1 - C(:,10)) \& (1 - C(:,11)) \& (1 - C(:,12)) \& (1 - RF(:,1))) \& (1 - (1 - (1 - C(:,13)) \& RF(:,2)) \& (1 - C(:,2)) \& C(:,13)))$	1
8	$1 - (1 - C(:,4)) \& (1 - C(:,5)) \& (1 - C(:,1) \& (1 - (1 - (1 - C(:,7))) \& (1 - C(:,8)) \& (1 - C(:,9)) \& (1 - C(:,10)) \& (1 - C(:,11)) \& (1 - C(:,12)) \& (1 - RF(:,1))) \& (1 - (1 - (1 - C(:,3)) \& (1 - RF(:,3))) \& (1 - C(:,6)) \& (1 - RF(:,4))) \& (1 - (1 - (1 - C(:,13)) \& (1 - RF(:,2)))$	1

“Event Tree and Fault Tree-type” Model -> “Uni-Modal Bound(UMB) Approach-type Mathematical Expression”